NER Annotation Guideline

Named Entity Recognition

BSAN 6200- Text Mining & Social Media Analytics

Assignment 4

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Version 3 [4/15/2026]

1. Purpose

The purpose of this annotation guideline is to establish rules for labeling named entities in the text to maintain consistency throughout labels. The quality of the NER model relies on consistent annotations.

2. Annotation Guide

Label the longest meaningful span of words. For example, “New York City” is labeled together and so is “University of California, Berkeley”

- Generic references like “the company” or “my mom’s friend” are not tagged as entities because they must be specific names.
- Titles such as “Dr.” are included the entity.
- Context is used for determining entity assignment
- Text in another language should be translated where possible and tagged appropriately as if it were in English
- Movie, book, tv show, and song titles are not considered one of the three entities used
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3. Annotation Definitions

Each entity type below includes a definition, correct examples, incorrect examples, and edge case notes. Entities in examples are marked with [brackets].

3.1 PERSON

Definition: Names of real people and fictional characters. Includes first names, last names, full names, twitter handles with full names, nicknames when used as a name, and titles when directly attached to a name.

Correct Extraction	Incorrect Extraction

Congrats to [jakclark95]	The famous [actress]- generic title
[Daniel James Evans] published a new paper	My [doctor] said to rest — generic role, not a named person
[Kane] loves coming through the ring	[Parents in law] are in town- generic relationship title; Not specific enough

⚠️ **Edge Cases:** When a name is ambiguous (e.g., 'Washington'), rely on context. If referring to a person, label as PERSON. If referring to a place, use GPE. If unclear, default to GPE for place names and PERSON only when clear personal reference exists. Twitter handles should be considered as an entity only if the actual name of the person is contained in the handle (ex. Jakclark95).

3.2 ORG

Definition: Companies, agencies, institutions, political parties, religious organizations, sports teams, and other named groups of people organized for a purpose.

Correct Extraction	Incorrect Extraction
If we can't beat the [Chargers]	The [church] in the city center
[Apple] released a new iPhone	All [hospitals] must comply — generic category, not named org
[Liverpool] played against [Man City]	The [government] announced — too vague without specific name

[Gateshead]'s media man just admitted to me	The [company's] media man- not specific

⚠️ **Edge Cases:** Abbreviations and acronyms (e.g., 'MIT', 'NASA', 'FBI') are valid ORG entities. When an organization name contains a location (e.g., 'University of California'), label the entire span as ORG, not as separate ORG + GPE entities. The word "Government" should be included as an org only if it is capitalized and is used as a proper noun. A band name such as "The Strokes" would be considered a organization.

3.3 LOC

Definition: All locations: countries, cities, mountain ranges, bodies of water, regions, continents, and other geographic features that are not governed political entities.

Correct Extraction	Incorrect Extraction
I'm having a baguette in [Paris]	Love the relaxed [coffee shop] — not a specific place
The [Rocky Mountains] extend through several states	Happy new year [world] – too general

⚠️ **Edge Cases:** Continents (e.g., 'Europe', 'Asia') are labeled as LOC. Neighborhoods and informal areas (e.g., 'Silicon Valley', 'Wall Street' as a location) can be LOC if used geographically, or ORG/GPE if used metonymically

4. Quick Reference Table

Use this table as a quick lookup during annotation.

Entity	What to Label	Example	NOT This
PERSON	Named individuals	Elon Musk	the CEO
ORG	Named organizations	Goldman Sachs	the bank
LOC	Geographic features	Pacific Ocean	My house

5. Annotation Process

5.1 Tools

We use Human Signal for annotation.

5.2 Workflow

1. Read the full sentence before annotating any entities.
2. Identify all named entities in the sentence.
3. For each entity, select the complete span (longest meaningful match).
4. Assign the appropriate label from this guideline.
5. If unsure, flag the sentence for discussion (add a comment in Human Label).
6. After completing a batch, review your annotations for consistency.

5.3 Quality Control

Inter-annotator agreement will be measured periodically. If two annotators disagree on more than 10% of entities in a shared batch, the team will discuss and update this guideline. Target: Cohen's Kappa ≥ 0.80.

6.0 Change Log

Version	Date	Changes
1.0	April 12, 2026	Initial version with 3 entity types based on spaCy label set.

1.1	April 14, 2026	First iteration where we noticed the kinds of edge cases and added them to the specifics of each entity type.
1.2	April 17, 2026	Second iteration where we used poor model performance to be even more specific with our labeling and put more difficult examples for each entity to serve as better guidance.